AMIRHOSSEIN KARIMI

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CAREER SUMMARY

Backend Developer | Python | Django | AWS Serverless

Backend Developer focused on Python, Django, and AWS serverless architectures. I have practical experience building containerized REST APIs and designing data processing pipelines that handle heavy workloads. In my recent roles, I focused on writing memory-efficient code to process multi-gigabyte geospatial files directly from S3, as well as automating complex data validation systems for clinical research. I care about writing tested, maintainable code and solving real infrastructure bottlenecks. Currently seeking a Junior Backend Developer role to build reliable and scalable systems.

TECHNICAL SKILLS

- **Languages & Databases:** Python, R, SQL, PostgreSQL, Redis
- **Frameworks & Cloud:** Django, Django REST Framework, Celery, Pandas, Matplotlib, rasterio, AWS (Lambda, S3)
- **DevOps & Tools:** Docker, Docker Compose, GitHub Actions, Pytest, unittest.mock, Git, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Software & Data Engineering Intern — Politecnico di Torino (Hospital Research Collaboration) | May 2026 – Nov 2026

- Built an automated Python and R pipeline that validates, scores, and categorizes patient questionnaire responses across 13+ clinical instruments (including PHQ-9 and PSQI).
- Wrote structural quality-control scripts to automatically scan incoming datasets for missing values, duplicate submissions, and out-of-range answers.
- Built an internal visualization module using Matplotlib to plot per-patient score trends and treatment timelines.

Python Developer Intern — StarView | Turin, Italy | Nov 2025 – Jan 2026

- Independently designed and implemented 3 serverless AWS Lambda functions as part of a 12-function geospatial data pipeline, processing multi-gigabyte .tif and ZIP files directly from S3 using Python, rasterio, and VSIS3 streaming.
- Engineered a custom S3SeekableFile class to stream large geospatial archives directly from S3, avoiding loading entire files into memory and substantially reducing Lambda memory overhead.
- Mitigated a Zip Slip path traversal vulnerability through independent research, and implemented lazy loading to reduce cold-start latency, with 100% unit test coverage via unittest.mock.

PROJECTS

Finance-Management API | Personal Project

- Architected a fully containerized personal finance REST API using Django REST Framework, PostgreSQL, and Docker Compose, featuring JWT authentication, full CRUD for transactions and budgets, and auto-generated Swagger/OpenAPI documentation.
- Implemented asynchronous budget alert notifications using Celery and Redis task queues, and established a complete CI/CD pipeline via GitHub Actions with Pytest test suite ensuring code reliability on every push.

E-Commerce REST API | Co-Developer

- Co-built a Django REST Framework e-commerce backend with PostgreSQL and Docker: JWT authentication, product catalog, shopping cart, and order management.
- Worked in a 2-person team using feature branches and conventional commits; documented the API with OpenAPI.

EDUCATION

MSc - Computer Engineering (Software specialization) | Politecnico di Torino, Turin, Italy | Sep 2026 – Present
BSc - Computer Engineering | Politecnico di Torino, Turin, Italy | Sep 2022 – Nov 2026

LANGUAGES

English (Fluent), Italian (Basic), Persian (Native)